

Traceability Metrics

Risk (Week7 and 8)	Control/Requirement (800-171r3)	Metric (from above)	Evidence Pointer (repo file/command)
Insufficient published metrics for certain PQC variants	03.02.01: Establish baseline of security-relevant data sources (“Establish and maintain... the set of information necessary to support mission/business processes.”; aligns w/ CSF GOV-01)	Data Source Diversity (# of unique sources cited / total cited); Coverage (# of industry sources reviewed)	docs/search_log.txt, docs/source_count.csv
Research availability introducing bias or incorrect results	03.05.01: Security Review of Research Content (“Review and validate the accuracy and relevance of information”)	Peer Review Coverage (% of findings peer reviewed or externally validated)	docs/peerreview_log.txt
Reliance on reviewed papers risking outdated/error propagation	03.02.03: Content Update and Validation (“Ensure information is current, accurate, regularly reviewed”)	Reference Freshness (avg. publication year/date in source log)	docs/source_log.csv
Privacy exposure from leakage of cryptographic	03.13.02: Information Transmitted Confidentiality (“Protect the	Privacy Incident Count (# of test runs with	src/paramleak_check.py, docs/logs/test_output.log

parameters/meta data	confidentiality of CUI during transmission”)	parameter leakage detected)	
Re-identification risk through linked data	03.13.01: Data De-Identification and Segmentation (“Separate and minimize CUI exposure to prevent linkage/re-identification”)	De-ID Enforcement Ratio (# of datasets de-identified / total datasets used)	docs/logs/data_deid.log
Over-collection of unnecessary metadata	03.04.03: Data Minimization/Usage Limitation (“Limit collection and retention to the minimum necessary”)	Metadata Minimization Rate (% of runs with only required fields logged)	src/meta_check.py, docs/logs/collection_output.log
Dual-use/misuse risk of research (possible system weakening)	03.01.01: Research Purpose Restriction (“Restrict research use to authorized/ethical purposes”; aligns with CSF GOV-06)	Purpose Declaration Completion (% of experimental scripts with explicit use statement)	docs/experiment_manifest.txt
Unsafe storage/sharing of sensitive crypto test data/keys	03.08.02: Data Storage Security (“Protect the confidentiality of CUI at rest and during sharing/storage/transfers”)	Secure Storage Compliance (# of test datasets / # stored per encryption policy)	src/storage_policy_check.py, docs/logs/storage_output.log